

Weihan Li

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Education

Georgia Institute of Technology, PhD in Machine Learning	Sept 2023 – Present
Georgia Institute of Technology, Master of Science in Computer Science	2023 – 2025
• Ph.D. Advisor: Prof. Anqi Wu	
Zhejiang University, Research Assistant	Sept 2020 – May 2023
• Advisor: Prof. Gang Pan	
Zhejiang University, Bachelor in Computer Science and Technology	Sept 2016 – June 2020

Research Summary

My current research is in **robot learning for dexterous bimanual manipulation**: how multi-fingered robot hands can learn diverse manipulation skills from human demonstrations, and how skills learned in simulation can transfer reliably to real hardware. Previously, I studied how multiple brain regions communicate and how animal behavior can be understood at scale, with applications to brain disorders and brain-computer interfaces.

Research Experience: Robot Learning

Dexterous Bimanual Manipulation from Human Demonstrations	Aug 2026 – Present
Platform: bimanual 20-DoF dexterous hands (Wuji, Sharpa) on a Vega humanoid	
• Turning human hand–object demonstrations into physically feasible robot demonstrations through kinematic and physics-based retargeting. • Learning residual reinforcement-learning policies on top of the retargeted demonstrations to perform contact-rich manipulation robustly. • Working toward sim-to-real transfer by aligning simulation with real-robot physics and evaluating policies against hardware-readiness criteria.	

Research Experience: Machine Learning for Neuroscience and Behavior

Multimodal Animal Behavior Analysis from Pose and Video	Sep 2025 – Jul 2026
• Proposed a pose–video fusion framework in which the quantization residual of a pose-based motion encoder identifies the frames that need visual context, so that video features are added only where pose is insufficient. • Applied the same model to supervised behavior classification and unsupervised behavior segmentation on three mouse datasets, outperforming pose- and video-based baselines on most tasks, with the largest gains in multi-animal social settings. • Published at NeurIPS 2026. • Co-developed BehaviorVLM, a finetuning-free approach to behavior understanding based on vision–language reasoning.	
MBRILA: An Open-Source GP State-Space Framework for Multi-Region Brain Communication	Oct 2025 – May 2026
• Developed a general platform and algorithmic framework for Gaussian-process state-space (GP-SSM) modeling, in which any Gaussian-process model can be converted into an equivalent state-space form and trained with scalable, linear- or parallel-time inference. • Provides shared building blocks (kernel-to-SSM conversion, multi-region latent dynamics, scalable inference) on which new GP-SSM methods can be built; MRM-GP and ADM are implemented as instances of the framework. • Released under the MIT license at github.com/BRAINML-GT/MBRILA.	

- Modeling Time-Varying Communication Across Brain Regions (ADM)** Aug 2024 – May 2025
- Proposed a state-space model that captures time-varying, directional interactions among multiple brain regions.
 - Established a general correspondence between Gaussian processes and equivalent state-space models, allowing flexible and interpretable priors on inter-regional communication.
 - Developed parallel-scan inference that reduces the time complexity from $O(T^3)$ to $O(\log T)$, enabling training on long multi-region recordings.
 - Published at ICML 2025 as an **Oral** presentation.
- Discovering Directional, Frequency-Specific Communication Across Brain Regions (MRM-GP)** Sep 2023 – Jan 2024
- Approximated multi-output Gaussian processes with state-space models to infer directional communication across brain regions in linear time.
 - Converted complex-valued spectral kernels into state-space dynamics, explicitly modeling frequency-specific coupling and phase delays between regions.
 - Recovered feedforward and feedback interactions consistent with known anatomy on multi-region recordings.
 - Published at ICML 2024.
- Online Neural Sequence Detection with Hierarchical Dirichlet Point Process** Jan 2022 – May 2022
- Developed a Bayesian nonparametric model that detects recurring neural sequences in spike trains and infers the number of sequence types from data.
 - Derived an online inference algorithm based on particle filtering, detecting sequences as spikes arrive and scaling to long recordings.
 - Published at NeurIPS 2022.

Publications

- Learning When Visual Context Matters for Mouse Behavior Analysis* NeurIPS 2026
 Weihan Li, Jingyang Ke, Qinheng Pu, Yule Wang, Chengrui Li, Anqi Wu
- BehaviorVLM: Unified Finetuning-Free Behavioral Understanding with Vision-Language Reasoning* preprint [\[link\]](#)
 Jingyang Ke*, Weihan Li*, Amartya Pradhan, Jeffrey Markowitz, Anqi Wu
 *Equal contribution; authors listed alphabetically.
- A Factorized Low-Rank RNN Framework for Uncovering Independent Neural Latent Dynamics and Connectivity* ICML 2026 Spotlight [\[link\]](#)
 Chengrui Li, Yunmiao Wang, Yule Wang, Weihan Li, Dieter Jaeger, Anqi Wu
- Uncovering Semantic Selectivity of Latent Groups in Higher Visual Cortex with Mutual Information-Guided Diffusion* ICLR 2026 [\[link\]](#)
 Yule Wang, Joseph Yu, Chengrui Li, Weihan Li, Dieter Jaeger, Anqi Wu
- Learning Time-Varying Multi-Region Brain Communications via Scalable Markovian Gaussian Processes* ICML 2025 Oral [\[link\]](#)
 Weihan Li, Yule Wang, Chengrui Li, Anqi Wu
- Exploring Behavior-Relevant and Disentangled Neural Dynamics with Generative Diffusion Models* NeurIPS 2024 [\[link\]](#)
 Yule Wang, Chengrui Li, Weihan Li, Anqi Wu
- Multi-Region Markovian Gaussian Process: An Efficient Method to Discover Directional Communications Across Multiple Brain Regions* ICML 2024 [\[link\]](#)
 Weihan Li, Chengrui Li, Yule Wang, Anqi Wu

- A Differentiable Partially Observable Generalized Linear Model with Forward-Backward Message Passing* ICML 2024 [\[link\]](#)
Chengrui Li, **Weihan Li**, Yule Wang, Anqi Wu
- Forward χ^2 Divergence Based Variational Importance Sampling* ICLR 2024 Spotlight [\[link\]](#)
Chengrui Li, Yule Wang, **Weihan Li**, Anqi Wu
- Online Neural Sequence Detection with Hierarchical Dirichlet Point Process* NeurIPS 2022 [\[link\]](#)
Weihan Li, Yu Qi, Gang Pan
- Efficient Point-Process Modeling of Spiking Neurons for Neuroprosthesis* EMBC 2021 [\[link\]](#)
Weihan Li, Cunle Qian, Yu Qi, Yiwen Wang, Yueming Wang, Gang Pan

Technical Skills

Robotics & Simulation: Isaac Lab, MuJoCo, Motion Retargeting, Sim-to-Real Transfer

Machine Learning: PyTorch, Reinforcement Learning, Vision-Language Models

Languages: Python, C++

Teaching and Academic Services

Teaching Experience: TA for Georgia Tech CSE-8803 Statistical Machine Learning Models for Neural and Behavioral Data Analysis.

Conference Reviewer: NeurIPS 2023/2024/2025/2026, ICML 2024/2025/2026, ICLR 2024/2025/2026.